Tilmicin Oral Solution 25% w/v

Composition

Tilmicosin phosphate equivalence to Tilmicosin ………………250 grams
Excipients q. s. to …………………………………………………1 L

Product Description

A yellow clear solution

Pharmacodynamics

Tilmicosin is a semi-synthetic antibiotic of the macrolide group and is believed to affect protein synthesis. It has bacteriostatic action but at high concentrations it may be bactericidal.

This antibacterial activity is predominantly against Gram-positive bacteria although some gram-negative bacteria and mycoplasmas of bovine, porcine, ovine and avian origin are also affected by this drug. In particular, its activity has been demonstrated against the following microorganisms:

- Pigs: *Mycoplasma hyopneumoniae, Pasteurella multocida* and *Actinobacillus pleuropneumoniae*
- Chickens: *Mycoplasma gallisepticum* and *Mycoplasma synoviae*
- Calves: *Mannheimia haemolytica, Pasteurella multocida, Mycoplasma bovis* and *M. dispar.*

Scientific evidence suggests that macrolides act synergistically with the host immune system. Macrolides appear to enhance phagocyte killing of bacteria. Tilmicosin has been shown to inhibit in vitro the replication of the Porcine Reproductive and Respiratory Syndrome virus in alveolar macrophages in a dose dependent fashion.

Cross-resistance between tilmicosin and other macrolides and lincomycin has been observed. Macrolides inhibit protein synthesis by reversibly binding to the 50S ribosomal subunit. Bacterial growth is inhibited by induction of the separation of peptidyl transfer RNA from the ribosome during the elongation phase.

Ribosomal methylase, encoded by the erm gene, can precipitate resistance to macrolides by alteration of the ribosomal binding site.

The gene that encodes for an efflux mechanism, mef, also brings about a moderate degree of resistance.

Resistance is also brought about by an efflux pump that actively rids the cells of the macrolide. This efflux pump is chromosomally mediated by genes referred to as acrAB genes.

Pharmacokinetic

Whilst blood concentrations of tilmicosin are low, there is pH-dependent macrophage accumulation of tilmicosin in inflamed tissues.

Pigs: After oral administration of 200 mg tilmicosin/l drinking water, the average active substance concentrations detected in lung tissue, alveolar macrophages and bronchial epithelium 5 days after the start of treatment were 1.44 μg/ml, 3.8 μg/ml and 7.4 μg/g respectively.

Poultry: As early as 6 hours after oral administration of 75 mg tilmicosin/l drinking water, the average active substance concentrations detected in lung and alveolar tissue were 0.63 μg/g and 0.30 μg/g respectively. 48 hours after the start of treatment, the tilmicosin concentrations in lung and alveolar tissue were 2.3 μg/g and 3.29 μg/g respectively.

Calves: As early as 6 hours after oral administration of 25 mg tilmicosin/kg body weight/day in milk replacer, an average active substance concentration of 3.1 μg/g was detected in lung tissue. 78 hours after the start of treatment, the tilmicosin concentration in lung tissue was 42.7 μg/g. Therapeutically effective concentrations of tilmicosin were measured up to 60 hours after treatment.

Indications for use, specifying the target species

Pigs: For the treatment and metaphylaxis of respiratory disease in pig herds, associated with *Mycoplasma hyopneumoniae, Pasteurella multocida, Actinobacillus pleuropneumoniae* and other organisms susceptible to tilmicosin.

Chickens: For the treatment and metaphylaxis of respiratory disease in chicken flocks, associated with *Mycoplasma gallisepticum* and *M. synoviae.*

Calves: For the treatment and metaphylaxis of bovine respiratory disease, associated with *Mannheimia haemolytica, Pasteurella multocida, Mycoplasma bovis, M. dispar* and other organisms susceptible to tilmicosin.

The presence of the disease in the group/flock must be established before the product is used.

Recommended Dose

For oral use only.

Pigs: To be included in the drinking water to provide a daily dose of 15-20 mg/kg bodyweight for 5 days, which may be achieved by the inclusion of 200 mg tilmicosin per litre (80 ml Tilmicin per 100 litres)?

Chickens (except hens producing eggs for human consumption): To be included in the drinking water at a daily dose of 15-20 mg/kg bodyweight in chickens for 3 days, which may achieved by the inclusion of 75 mg tilmicosin per litre (30 ml Tilmicin per 100 litres).

Calves: To be included in milk replacer only, at a dose of 12.5 mg/kg bodyweight and given twice daily for 3-5 consecutive days, which may be achieved by the inclusion of 1 ml of product every 20 kg bodyweight. One 240 ml bottle of Tilmicin is sufficient to medicate 300 litres of drinking water for pigs or 800 litres of drinking water for chickens.

One 240 ml bottle of Tilmicin are sufficient to medicate in milk replacer respectively 12 to 20 veal calves each of 40 kg bodyweight depending on the duration of treatment.

To ensure a correct dosage, bodyweight should be determined as accurately as possible to avoid under-dosing.

The required dose should be measured using suitably calibrated measuring equipment.

Only sufficient medicated drinking water should be prepared to cover the daily requirements.

Route of Administration

Oral

Contraindications

Do not allow horses and other equines access to drinking water containing tilmicosin.
Do not use in case of hypersensitivity to tilmicosin or to any of the excipients.

Warnings

Important: Must be diluted before administration to animals.

Pigs and chickens: Water consumption should be monitored in order to guarantee adequate dosing. In case water consumption does not match quantities for which recommended concentrations were calculated, the concentration of Tilmicin has to be adapted in a way that the recommended dosage will be taken up by the animals or different medication should be considered.

Precautions

I. Special precautions for use in animals

For oral use only. Contains disodium edetate; do not inject

Severely ill individuals tend to drink less and may need simultaneous treatment, preferably by parenteral medication.

Inappropriate use of the product may increase the prevalence of bacteria resistant to tilmicosin and may decrease the effectiveness of treatment with tilmicosin-related substances.

The use of the product should be based on susceptibility testing of the bacteria isolated from the animal. If this is not possible, therapy should be based on local (regional, farm level) epidemiological information about susceptibility of the target bacteria.

Medicated drinking water or milk replacer should be prepared fresh every 24 hours.

Official, national and regional antimicrobial policies should be taken into account when the product is used.

II. Special precautions to be taken by the person administering the veterinary medicinal product to animals

- Tilmicosin may induce irritation. Macrolides, such as tilmicosin, may also cause hypersensitivity (allergy) followinginjection, inhalation, ingestion or contact with skin or eye. Hypersensitivity to tilmicosin may lead to cross reactions toother macrolides and vice versa. Allergic reactions to these substances may occasionally be serious and thereforedirect contact should be avoided.

- To avoid exposure during preparation of the medicated drinking water, wear overalls, safety glasses, andimpervious gloves. Do not eat, drink or smoke when handling this product. Wash hands after use.

- In the case of accidental ingestion, wash out mouth immediately with water and seek medical advice. In the eventof accidental skin contact, wash thoroughly with soap and water. In case of accidental eye contact, flush the eyeswith plenty of clean, running water.

- Do not handle the product if you are allergic to ingredients in the product.

- If you develop symptoms following exposure, such as skin rash, you should seek medical advice and show thephysician this warning. Swelling of the face, lips and eyes or difficulty in breathing are more serious symptoms andrequire urgent medical attention.

Interactions with Other Medicaments

None known.

Pregnancy and Lactation

The safety of tilmicosin has not been established during pregnancy, lactation or lay. Use only is according to the benefit/risk assessment by the responsible veterinary surgeon.

Side Effects

In very rare cases (less than 1 animal in 10,000 animals treated, including isolated reports), a decrease in water intake has been observed.

Symptoms and Treatment of Overdose

When pigs are offered drinking water containing 300 or 400 mg tilmicosin /litre (equivalent to 22.5-40 mg tilmicosin /kg bodyweight or 1.5-2 times the recommended concentration) commonly animals exhibit a reduced water intake. Although this has a self-limiting effect on tilmicosin intake, it could, in extreme circumstances, result in dehydration. This can be corrected by the removal of the medicated drinking water and replacement with fresh unmedicated water.

No symptoms of overdose have been seen in chickens given drinking water containing levels of tilmicosin up to 375 mg/litre (equivalent to 75-100 mg tilmicosin /kg bodyweight or 5 times the recommended dose) for 5 days; daily treatment with 75 mg tilmicosin /litre (equivalent to the maximum recommended dose) for 10 days resulted in a reduction in faecal consistency.

No symptoms of overdose, with exception of a slight decrease in the milk consumption, have been seen in calves given twice daily doses 5 times the maximum recommended dose or for twice the maximum recommended duration of treatment.

Withdrawal Period

Pigs: meat and offal: 14 days
Chickens: meat and offal: 12 days
Calves: meat and offal: 42 days

Not authorised for use in animals producing milk for human consumption.
Not authorised for use in birds producing eggs for human consumption.
Do not use within 2 weeks of the start of the laying period.

Storage Condition

Do not store above 30°C and protected from light.

Packaging

White plastic bottle 240 ml
White plastic bottle 1L

Manufacturer

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Marketing Authorization Holder

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